

Probability & Simulation

INS-605: Data Analysis II



Content

1 Introduction & Motivation

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4 Simulation Methods

Helpful Resources

Introduction to Probability for Data Science, Stanley Chan.

Probability for **Data Science**

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Introduction to Probability for Data Science

Second Edition

by **Stanley Chan** · Michigan Publishing, 2026

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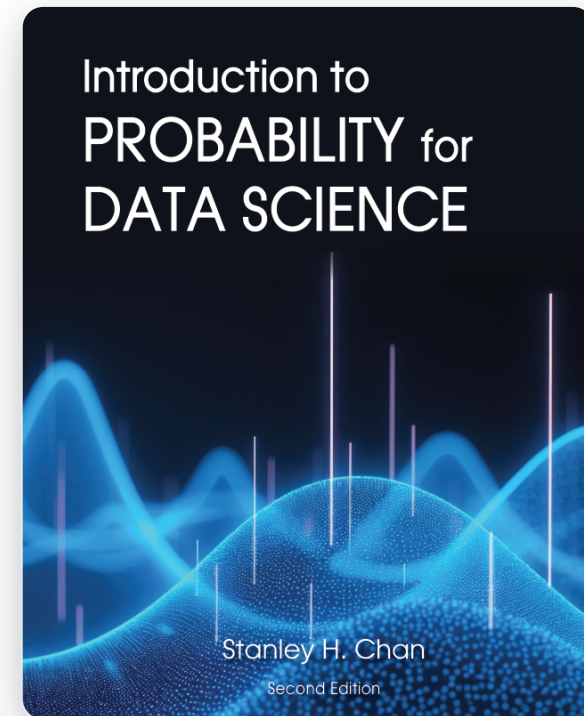
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@book{chan2026probability,  
  title   = {Introduction to Probability for Data Science},  
  author  = {Chan, Stanley H.},  
  year    = {2026}.
```

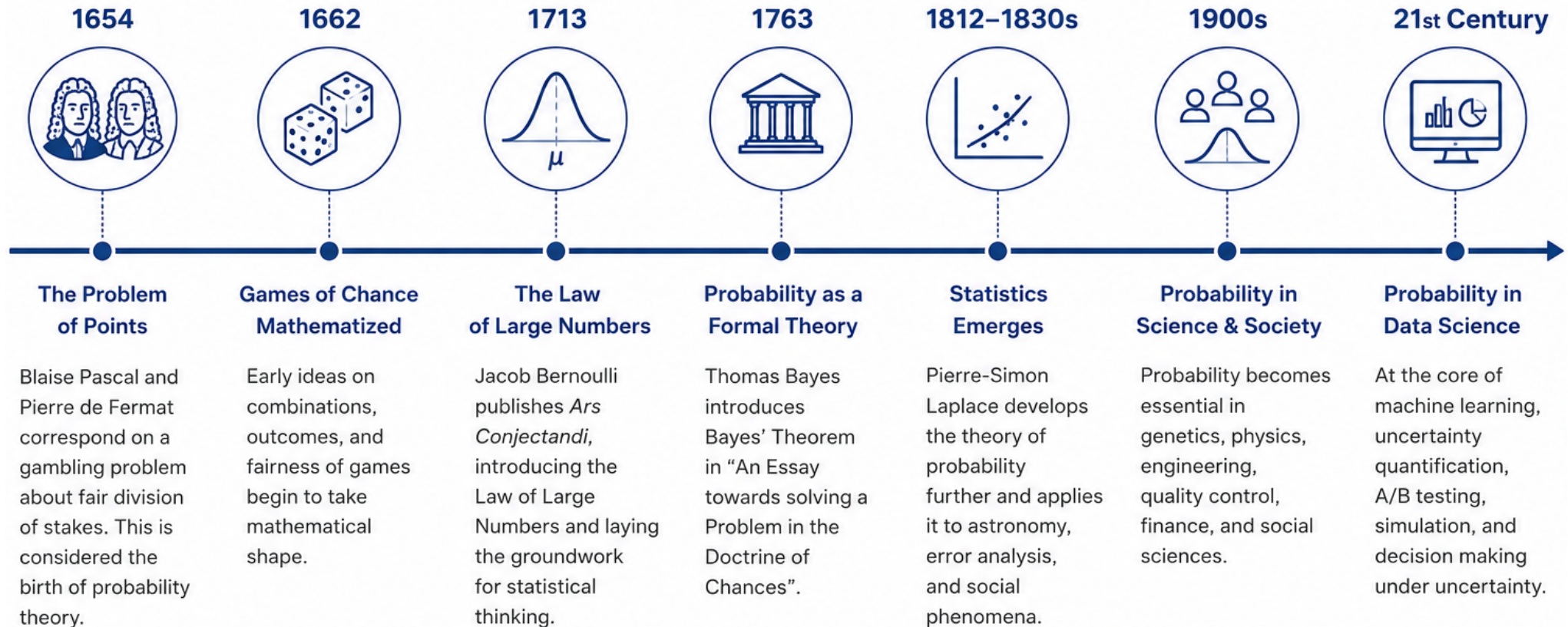


1 Introduction & Motivation

1.1 History

The Birth of Probability

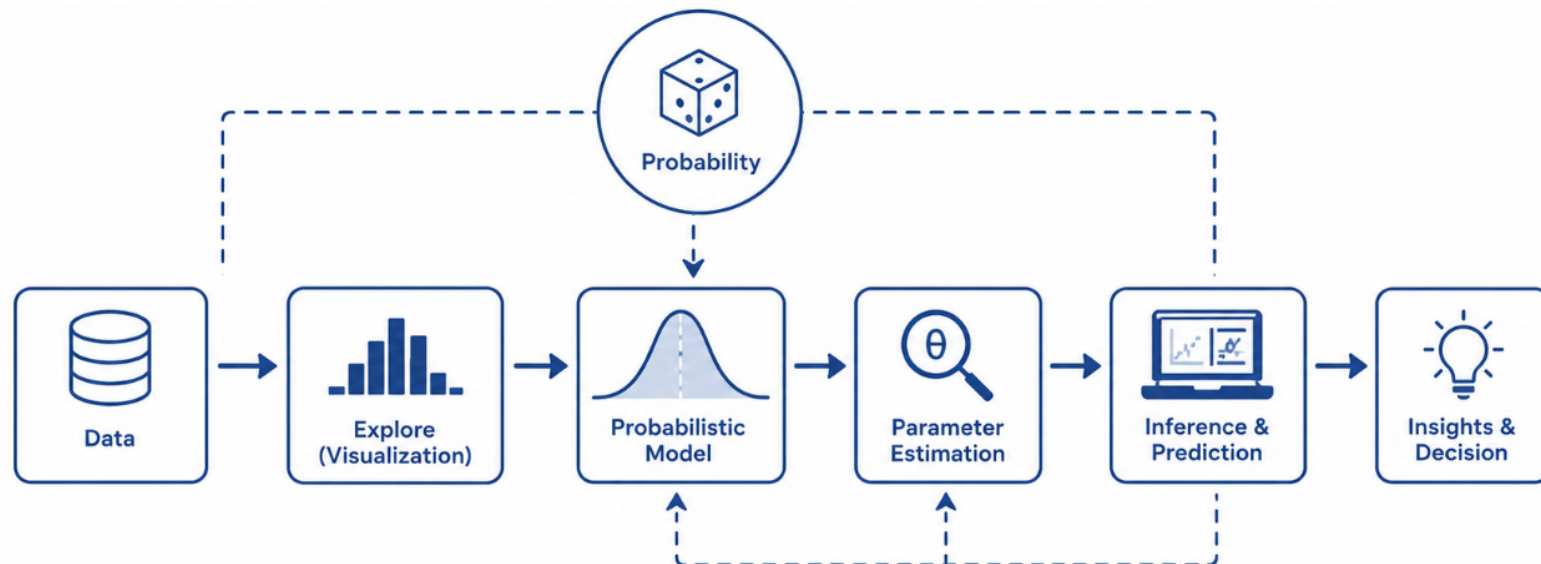
From games of chance to the foundation of modern data science.



- **Our focus:** Probability in Data Science & Simulation.

1.2 Probability In Data Science

- In Data Science, probability is important in various parts:
 - **Sampling & Data Collection:** Gathering unbiased, representative data.
 - **Anomaly Detection:** Spotting rare events and unexpected outliers.
 - **Hypothesis Testing:** Proving whether observed patterns are real or random chance.
 - **Unsupervised Clustering:** Grouping similar data based on density and likelihood.
 - **Predictive & Generative Models:** Forecasting outcomes and synthesizing new data.

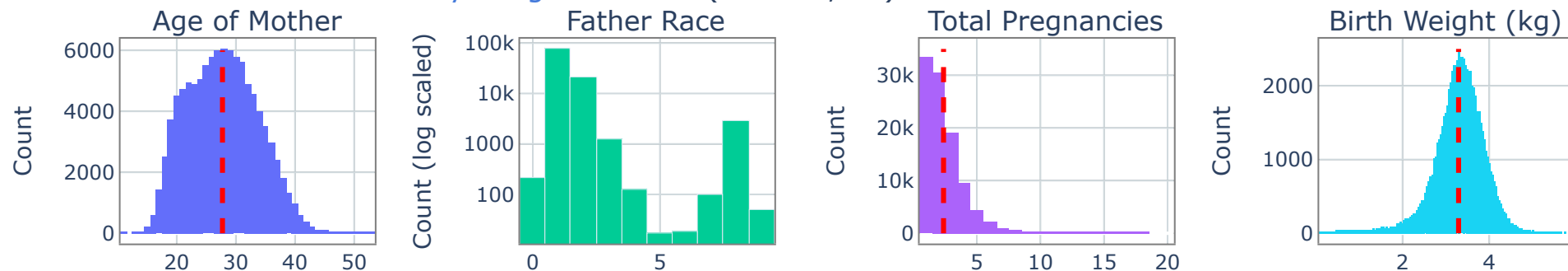


1.3 Motivation

- **Signal vs. Noise:** Real-world data seamlessly blends *meaningful patterns* with *inherent uncertainty*.
- **Avoid Misleading Insights:** To extract *useful signal*, we must quantify *noisy components*.
- **Characterize Uncertainty:** The first step toward reliable analysis is understanding how *uncertainty behaves*.
- **Role of Probability: Probability theory** serves as a foundational tool to *quantify* and *manage* this randomness.

► Code

Some Variables from **Baby Weight Dataset** (101400, 37)



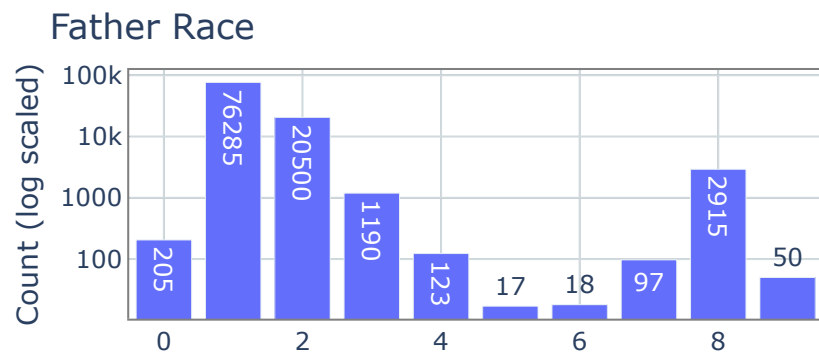
2 Probability Distributions & Models

2.1 Discrete Random Variables

Review

- To describe data mathematically, we use **Random Variables** denoted by $X, Y, Z, \dots$
- Unlike **data type**, we classify RV into two different types:
 - **Discrete V.R.**, $X \in \mathcal{S} \subset \mathbb{N} = \{1, 2, 3, \dots\}$ ($\mathcal{S}$: sample space.)
 - **Continuous V.R.**, $X \in \mathcal{S} \in \mathbb{R}$.

► Code



- If X : Race of Father,
 - $\mathcal{S} = \{0, 1, \dots, 9\}$.
- If a father is chosen at random, how likely that his race is '4'?

Probability Mass Function (PMF)

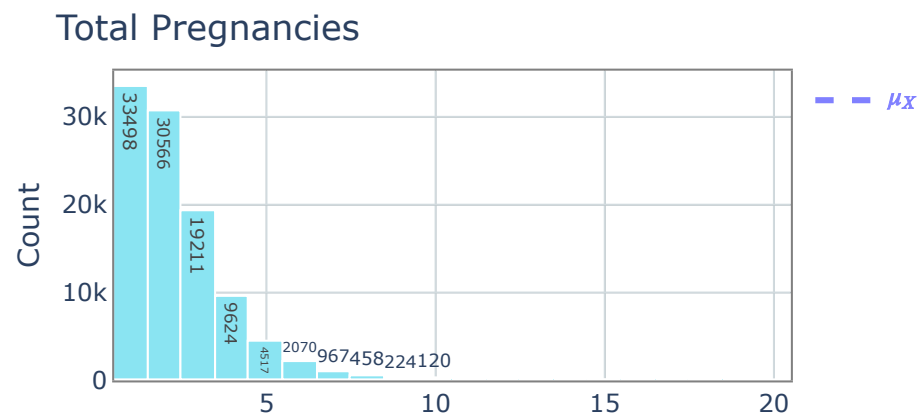
- If Y : Number of Pregnancies,

- $S_Y = \{1, 2, 3, \dots\}$
- $\hat{\mathbb{P}}(Y = 4) \approx \frac{9624}{101400} = 0.09$.

	proportion
TOTALP	
1	0.330355
2	0.301440
3	0.189458
4	0.094911
5	0.044546
6	0.020414
7	0.009536
8	0.004517
9	0.002209
10	0.001183

	proportion
TOTALP	
11	0.000700
12	0.000394
13	0.000128
14	0.000069
17	0.000039
15	0.000039
18	0.000020
16	0.000020
20	0.000010
19	0.000010

► Code



- **Probability Mass Function** f :

$$f(x_i) = p_i = \mathbb{P}(X = x_i), \forall x_i \in \mathcal{S}$$

- It describes **how likely** each possible outcome x_i is in a random experiment involving X .
- **Roughly:** We can approximate *usual* or *unlikely* outcomes and other characteristics of X .

Numerical Summary Of RV

- **Expectation** of a DRV X is the **mean/average** over the outcomes x_i 's weighted by their corresponding chances p_i 's, .i.e.,

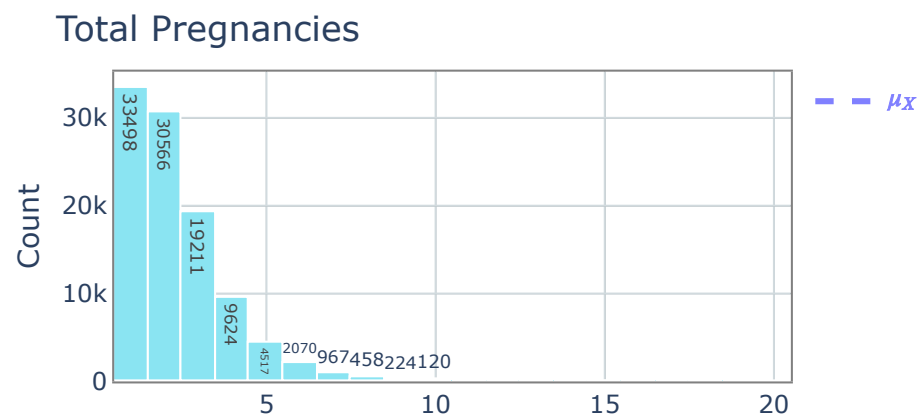
$$\mu_X = \mathbb{E}(X) = \sum_{i \geq 1} x_i p_i = x_1 p_1 + x_2 p_2 + \dots$$

- **Note:** Mean/average of real data is realized with $p_i = \hat{p}_i = \frac{n_i}{n}$.
- Ex: **Total Pregnancies**, $\bar{X}_n = 2.38$.

-
- **Variance** of a DRV X is the **average squared gaps** between outcomes x_i 's and the mean μ_X weighted by p_i 's, .i.e.,

$$\sigma_X^2 = \mathbb{V}(X) = \mathbb{E}[(X - \mu_X)^2] = \sum_{i \geq 1} (x_i - \mu_X)^2 p_i.$$

► Code

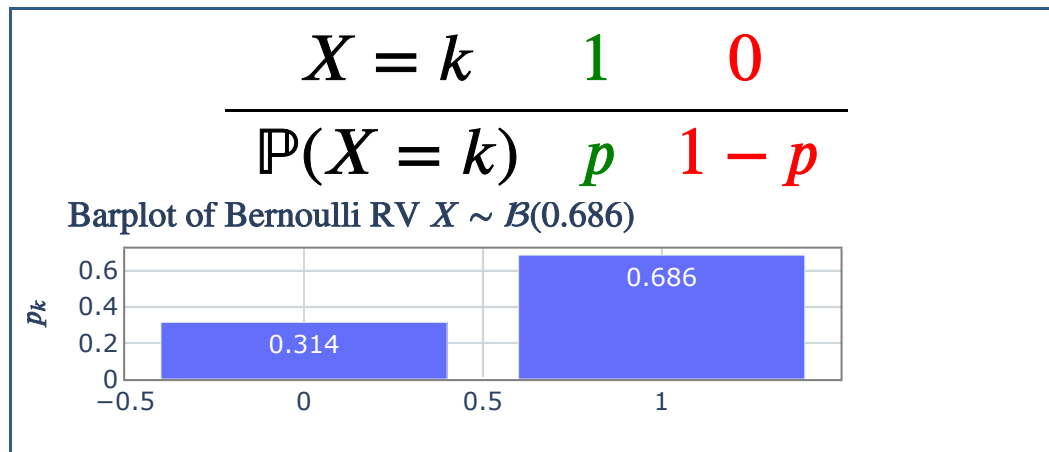


- **Large variance** indicates the *wider dispersion of outcomes* or *more uncertainty* of the RV
- Smaller **variance** indicates *lesser uncertainty*.
- **Standard Deviation (SD)** is more common to use as it is in the same units as the R.V: $\sigma_X = \sqrt{\mathbb{V}(X)}$.

Some Common DRVs

Bernoulli Distribution

- For *binary data* with only two possible outcomes {1 : ‘success’, 0 : ‘failure’}, the **Bernoulli Distribution** is used.
- Ex: Gender, Yes/No, Sick/Healthy, etc.
- Let $X \in \mathcal{S} = \{0, 1\}$ be such a data, denoted by $X \sim \mathcal{B}(p)$, where p is the probability of success, i.e., $p = \mathbb{P}(X = 1)$.
- PMF of X is given by: $\mathbb{P}(X = 1) = p$ and $\mathbb{P}(X = 0) = 1 - p = q$.



- If $X \sim \mathcal{B}(p)$, then
 - $\mathbb{E}[X] = p$
 - $\text{Var}(X) = p(1 - p)$
 - $\text{SD}(X) = \sqrt{p(1 - p)}$.

Some Common DRVs (Cont.)

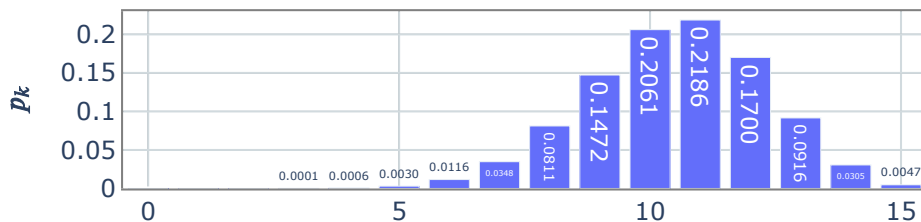
Binomial Distribution

- **The Number of Total Successes** occurs in a series of **n** independent **Bernoulli** trials yield a **Binomial** distribution, $X \sim \text{Bin}(n, p)$.
- Ex: The number of sick children in room of 25 newborns...
- If $X \sim \text{Bin}(n, p)$, then $S = \{0, 1, \dots, n\}$ and its **PMF** is given by:

$$P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k} \text{ for } k = 0, \dots, n.$$

$X = k$	0	1	2	...	$n - 1$	n
$\mathbb{P}(X = k)$	$(1 - p)^n$	$np(1 - p)^{n-1}$	$\frac{n(n-1)}{2!} p^2 (1 - p)^{n-2}$	...	$np^{n-1} (1 - p)$	p^n

Barplot of Binomial RV $X \sim \text{Bin}(13, 0.7)$



- If $X \sim \text{Bin}(n, p)$, then
 - $\mathbb{E}[X] = np$
 - $\text{Var}(X) = np(1 - p)$
 - $\text{SD}(X) = \sqrt{np(1 - p)}$.

Some Common DRVs (Cont.)

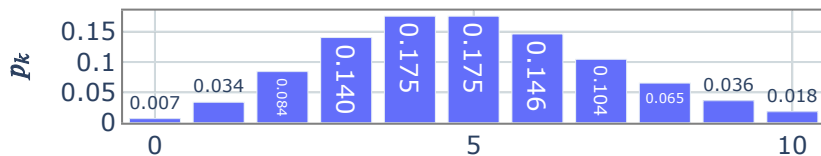
Poisson Distribution

- A **Poisson** RV X counts the number of (rare) events in a fixed interval of time or space, $X \sim \mathcal{P}(\lambda)$, $\lambda > 0$.
- Ex: *Number of wrong money transfers in an app per day*, or *Number of Total Pregnancies* a woman may have...
- If $X \sim \mathcal{P}(\lambda)$, then $\mathcal{S} = \{0, 1, 2, 3, \dots\}$ and its PMF is given by:

$$P(X = k) = e^{-\lambda} \frac{\lambda^k}{k!} \text{ for } k = 0, \dots, n.$$

$X = k$	0	1	2	3	4	...
$\mathbb{P}(X = k)$	$e^{-\lambda}$	$e^{-\lambda} \lambda$	$e^{-\lambda} \frac{\lambda^2}{2!}$	$e^{-\lambda} \frac{\lambda^3}{3!}$	$e^{-\lambda} \frac{\lambda^4}{4!}$	...

Graph of Poisson RV $X \sim \mathcal{P}(5)$



- If $X \sim \text{Bin}(n, p)$, then
 - $\mathbb{E}[X] = \text{Var}(X) = \lambda$
 - $\text{SD}(X) = \sqrt{\lambda}$.

Expectation, Variance & SD

i Note

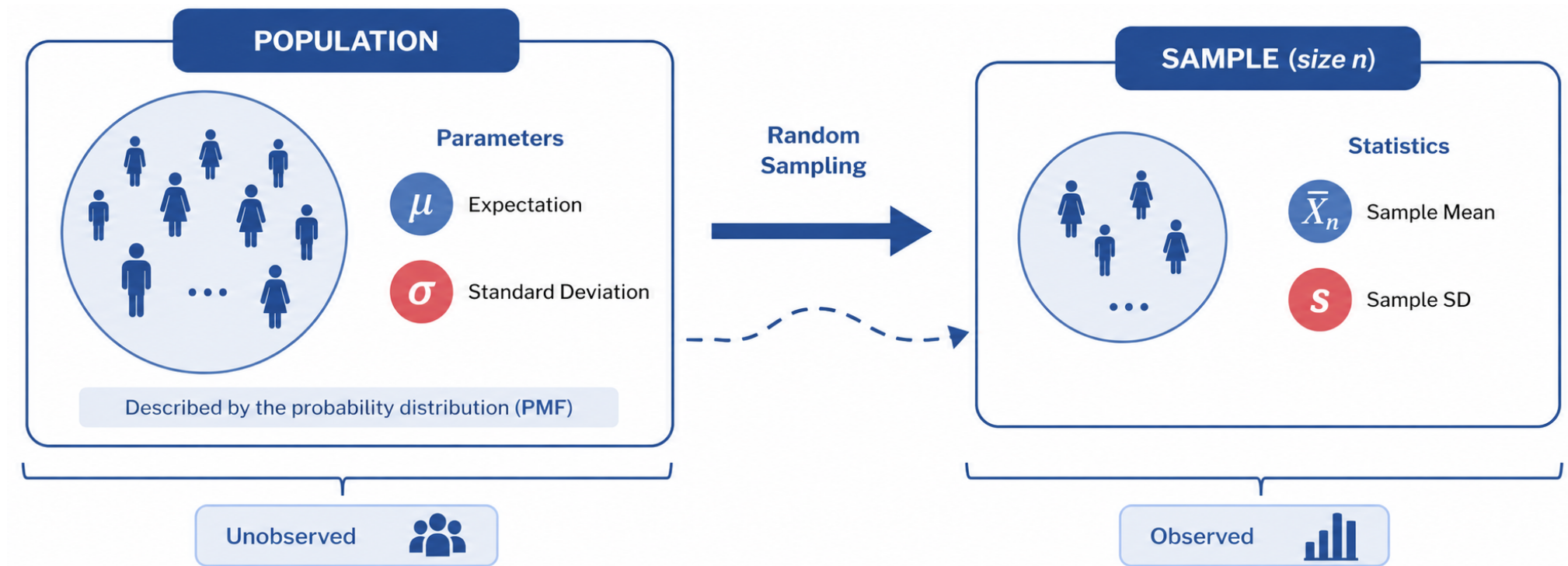
Theoretical Concepts

- **Expectation:** $\mu_X = \sum_{i \geq 1} x_i p_i$.
- **Variance:** $\sigma_X^2 = \sum_{i \geq 1} (x_i - \mu_X)^2 p_i$.
- **SD:** $\sigma_X = \sqrt{\sigma_X^2}$.
- Theoretical concepts.
- Computed from a probability distribution (**PMF**).
- Never be computed from data.
- They are called **parameters** of the *population*.
- Example:
 - Your true weight/height.
 - True average birth weight in Cambodia.
 - **Coefficient of Variation** (σ/μ) in Cambodia.

Empirical Concepts

- **S. Mean:** $\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$.
- **S. Variance:** $s_X^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X}_n)^2$.
- **S. SD:** $s_X = \sqrt{s_X^2}$.
- Empirical concepts.
- Computed from data.
- Used to estimate theoretical parameters.
- They are called **statistics** of the *sample*.
- Example:
 - Average across several scales.
 - Average over several babies.
 - Compute $\hat{\sigma}_n / \bar{X}_n$ on some sample.

Expectation, Variance & SD



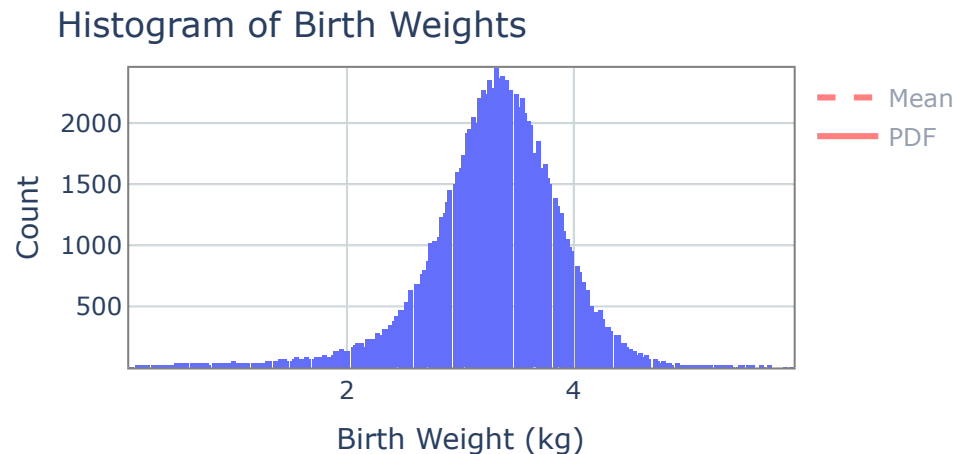
- There are two different beliefs in **parameter** of the population:
 - **Frequentist:** A parameter is a **fixed, unknown constant**.
 - **Bayesian:** A parameter is a **RV** with some distribution.

2.2 Continuous Random Variables

Review

- For **Ratio** measurements such as *weight*, *height*... the **sample space** is the set of all possible **real numbers**, i.e., $S = \mathbb{R}$.
- Each **outcome** can take any **real number** (we are seriously talking about a **precise** value among **uncountably infinite** possibilities).

► Code



- If X is the Birth Weight (kg),
- ‘**How likely to observe $X = 3\text{kg}$?**’, no longer possible!
- We can ask ‘**How likely is it that X takes a value between 2 and 3kg?**’.
- That’s $\mathbb{P}(2.9 \leq X \leq 3)$, but how do you compute it from the data?

Probability Density Function

- **Probability Density Function (PDF)** f of a CRV X **governs how likely** it is for X to take on any value.
- **PDF** satisfies the following properties:
 - $f(x) \geq 0$ for all $x \in \mathcal{S}$.
 - $\int_{-\infty}^{\infty} f(x)dx = 1$.

Warning

- Don't confuse PDF $f(x)$ with probability of $X = x$, which is 0 for continuous random variables.
- Probability of X being in a small interval $(x, x + \Delta x)$ is approximately $f(x)\Delta x$, which is the **area under the PDF curve over that interval**.

Numerical Summary Of CRVs

- For a CRV X with PDF f , one has:
 - $\mathbb{P}(a \leq X \leq b) = \int_a^b f(x)dx, \forall a \leq b.$

- Expectation** of a CRV X is defined by,

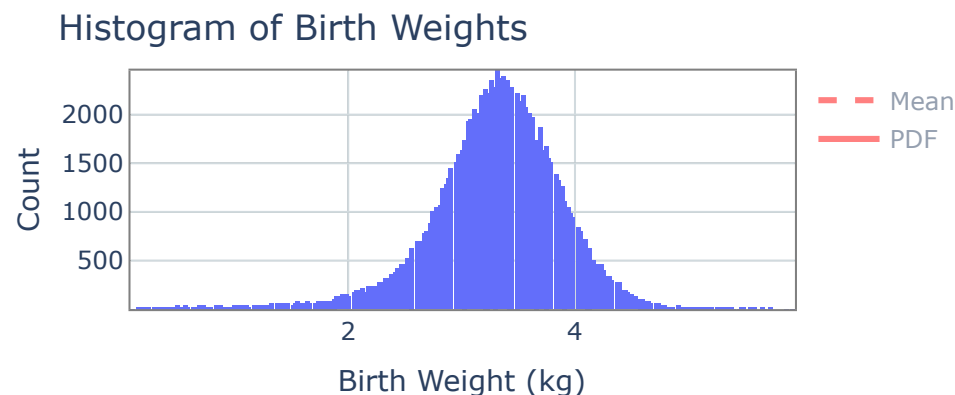
$$\mu_X = \mathbb{E}(X) = \int_{-\infty}^{\infty} x f(x) dx$$

- Just like in DRV case, μ_X is the **average** over the range of all possible x weighted by the density $f(x)$.

-
- Variance** of a CRV X is the **average squared gaps** between the range of all possible x and the mean μ_X weighted by $f(x)$, i.e.,

$$\sigma_X^2 = \mathbb{V}(X) = \mathbb{E}[(X - \mu_X)^2] = \int_{-\infty}^{\infty} (x - \mu_X)^2 f(x) dx.$$

► Code



- From the graph above, we can **estimate** and obtain the following density for **Birth weight** distribution:

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

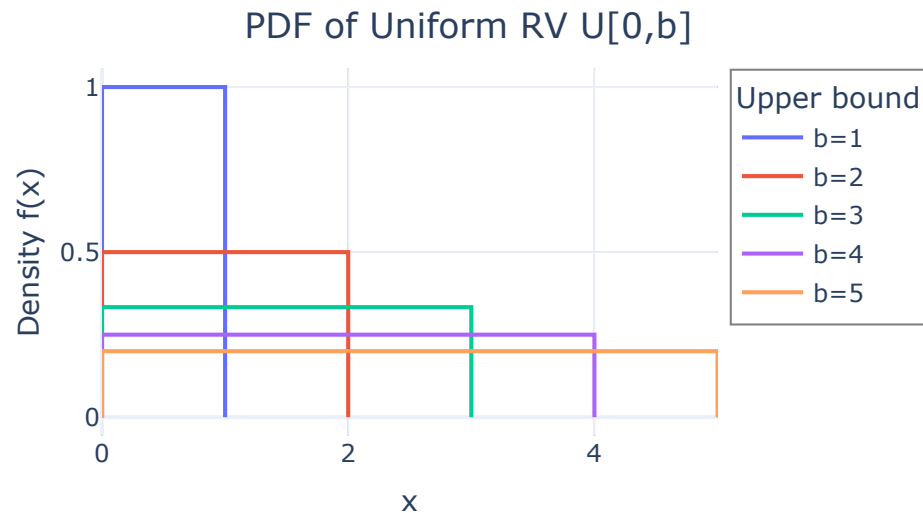
with $\sigma \approx 0.467$ and $\mu \approx 3.292$ (How 🤔?).

- Then, $\mathbb{P}(2 \leq X \leq 3) = \int_2^3 f(x) dx \approx 0.263.$
- And $\mathbb{P}(X \leq 2) = \int_{-\infty}^2 f(x) dx \approx 0.00285.$

Some Common CRVs

Uniform Distribution

- A RV U is called **Uniform** RV over an interval $[a, b]$ if it's equally likely to take any value in that interval, denoted by $U \sim \mathcal{U}[a, b]$.
- The **PDF** of a uniform RV: $f(x) = \begin{cases} \frac{1}{b-a}, & \text{for } x \in [a, b] \\ 0, & \text{otherwise} \end{cases}$.



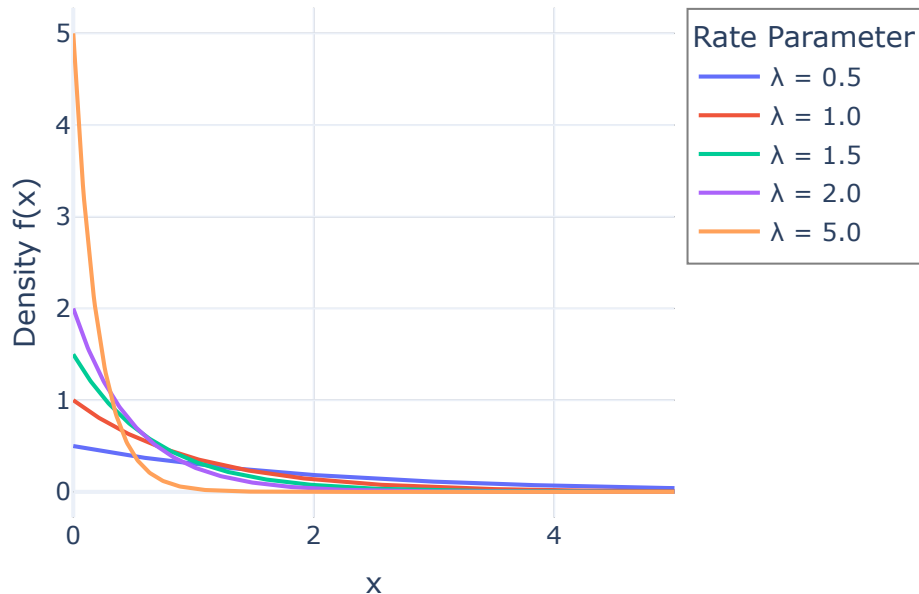
- If $U \sim \mathcal{U}[a, b]$, then
 - $\mathbb{E}(U) = \frac{a+b}{2}$
 - $\mathbb{V}(U) = \frac{(b-a)^2}{12}$
 - $\sigma_U = \sqrt{\frac{(b-a)^2}{12}}$.
- In computer, almost all random numbers are generated from $U[0, 1]$.

Some Common CRVs (Cont.)

Exponential Distribution

- A **positive** CRV is **Exponentially distributed** ($X \sim \mathcal{E}(\lambda)$) if its **PDF** is given by $f(x) = \lambda e^{-\lambda x}$, for some $\lambda > 0$ and $x \geq 0$.
- Ex: Waiting time for a customer call, arrival time of a bus...

Exponential Probability Density Functions (PDFs)



- If $X \sim \mathcal{E}(\lambda)$, then
 - $\mathbb{E}(X) = \sigma_X = \frac{1}{\lambda}$
 - $\mathbb{V}(X) = \frac{1}{\lambda^2}$
- It's used in many studies related to:
 - **Queueing theory** (e.g., time between customers)
 - **Birth and death processes...**

Some Common CRVs (Cont.)

Normal/Gaussian Random Variable

- A RV **normal** or **Gaussian** with mean μ and variance σ^2 is denoted by $X \sim \mathcal{N}(\mu, \sigma^2)$.
- For modeling natural measurements:
 - Heights, weights, sizes...
 - Measurement errors...
- Its **PDF** is defined for all $x \in \mathbb{R}$ by

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-(x-\mu)^2/(2\sigma^2)}.$$

- We have: $\mathbb{E}(X) = \mu$ and $\mathbb{V}(X) = \sigma^2$.

Important Properties & Inequalities

1. **Normality is preserved under linear transformations:**

$$\text{If } \begin{cases} X_1 \sim \mathcal{N}(\mu_1, \sigma_1^2) \\ X_2 \sim \mathcal{N}(\mu_2, \sigma_2^2) \\ X_1 \perp X_2 \end{cases} \Rightarrow Y = a_1 X_1 + a_2 X_2 \sim \mathcal{N}(a_1 \mu_1 + a_2 \mu_2, a_1^2 \sigma_1^2 + a_2^2 \sigma_2^2).$$

2. **Markov inequality:** For any RV $X > 0$ with $\mathbb{E}(X) < \infty$ and $a > 0$,

$$\mathbb{P}(X \geq a) \leq \frac{E(X)}{a}.$$

3. **Chebyshev's inequality:** For any RV X with $\mathbb{E}(X) = \mu < \infty$ and $\mathbb{E}(X^2) < \infty$ (or $\sigma^2 < \infty$), and $k \geq 1$,

$$\mathbb{P}(|X - \mu| \geq k\sigma) \leq \frac{1}{k^2}.$$

Main Theories

Law of Large Number (LLN)

- If $X_1, X_2, \dots, X_n$ is a sample of n i.i.d copies of a RV X (think about it like collecting similar data from the same source as X), assume that $\mathbb{E}(X) = \mu_X < \infty$ and let $\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$ be the sample mean of the sample, then one has:

- **Weak LLN:** $\left(\bar{X}_n \xrightarrow{P} \mu_X \text{ as } n \rightarrow \infty \right)$, i.e., for any $\epsilon > 0$ fixed, we have

$$\mathbb{P} \left(\left\{ \omega : \left| \bar{X}_n(\omega) - \mu_X(\omega) \right| > \epsilon \right\} \right) \rightarrow 0 \text{ as } n \rightarrow \infty.$$

- **Strong LLN:** $\left(\bar{X}_n \xrightarrow{a.s.} \mu_X \text{ as } n \rightarrow \infty \right)$, i.e.,

$$\mathbb{P} \left(\left\{ \omega : \lim_{n \rightarrow \infty} \bar{X}_n(\omega) = \mu_X(\omega) \right\} \right) = 1.$$

- This is the main theory that guarantees that the sample mean converges to the expected value of the random variable (mean of population).

Main Theories

💡 Central Limit Theorem

- With the same setting as in [LLN](#), and let $\sigma_X = \sqrt{\mathbb{V}(X)} < \infty$ be the standard deviation of X :

$$Z = \left(\frac{\overline{X}_n - \mu_X}{\sigma} \right) \xrightarrow{d} \mathcal{N}(0, 1) \text{ as } n \rightarrow \infty.$$



LAB STATISTICS CALCULATOR TUTORIAL BOOK

Central Limit Theorem

Explore the Central Limit Theorem with three interactive tools: Galton Board, Dice Sum Simulator, and Sample Mean Lab. Adjust sample sizes, trials, and source distributions, and view real-time histograms and density overlays. Compare empirical results with theoretical normal curves to observe convergence and variability.



3 Parameter Estimation

3.2 PMF/PDF To Likelihood

- Example: if newborn weights are $\mathcal{N}(\mu, 1)$, with known $\sigma = 1$ (kg) and unknown mean μ , for a baby with weight $X_1 = 3.2$ kg chosen from this population, which of the following means are most likely the **true underlying μ** of the population: 2.0, 3.1, 3.9 kg?
-
- Likelihood is the density of observing $X_1 = 3.2$ given a parameter μ :
 $L(\mu) = f(3.2|\mu)$.
- It's a function of μ .
- Parameter μ that maximizes $L(\mu)$ fits the data best.

Likelihood Function

- Using the visualization, we can choose a suitable family of distribution for the data that depends on a parameter θ .
- Then, for a fixed set of n i.i.d. observations $\{x_1, x_2, \dots, x_n\}$ drawn from this population, the likelihood function is defined as:

$$L(\theta) = f(x_1, x_2, \dots, x_n | \theta) = \prod_{i=1}^n f(x_i | \theta) \geq 0$$

- Maximum Likelihood Estimator $\hat{\theta}$ is the value that maximizes $L(\theta)$,

$$L(\hat{\theta}) = \max_{\theta} L(\theta).$$

Log-Likelihood Function

- Likelihood function is often difficult to work with as it's a product of probabilities and can be very small.
- **Log-likelihood** is an easier objective function to be maximized in order to find MLE:

$$\ell(\theta) = \log L(\theta) = \sum_{i=1}^n \log f(x_i | \theta).$$

- Maximum Likelihood Estimator $\hat{\theta}$ also maximizes $\ell(\theta)$,

$$\ell(\hat{\theta}) = \max_{\theta} \ell(\theta).$$

Summary

Why do we need to study the distribution of data?



It allows us to **quantify uncertainty** associated to the data.



It helps us understand the **underlying structure** of the data.



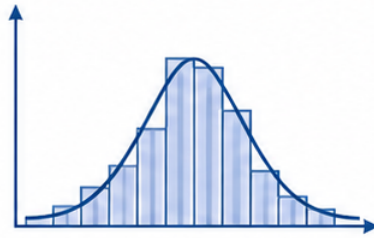
It enables us to make **predictions and inferences** based on the data.

1 Visualize the Data



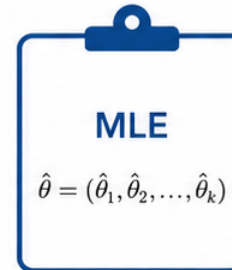
We begin by visualizing the data.

2 Choose a Distribution



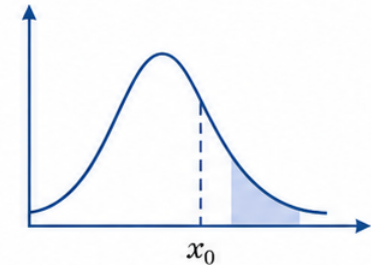
Then choose a distribution that best fits the data.

3 Estimate Parameters



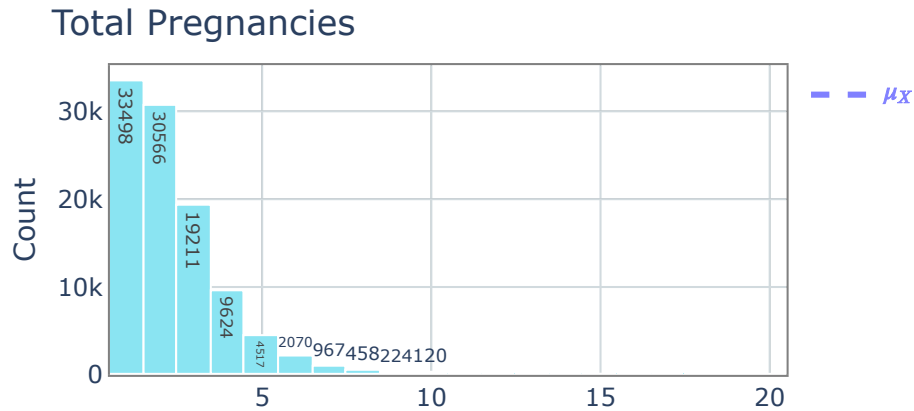
We then estimate the parameters of the chosen distribution using **Maximum Likelihood Estimation**.

4 Make Predictions and Inferences



We can then use the estimated parameters to make predictions and inferences about the data.

Application: Total Pregnancies



- What's probability distribution is suitable for **Total Pregnancies**?
- **Model:** $X \sim \mathcal{P}(\lambda)$ for some $\lambda > 0$.
- **Data:** Assume $x_1, \dots, x_n$ are from the above population and independent.
- **Log-likelihood:**
 $\ell(\lambda) = \sum_{k=1}^n [-\lambda + x_k \log(\lambda) - \log(x_k!)]$.

- **Optimization:**

$$\frac{d\ell(\lambda)}{d\lambda} = \sum_{k=1}^n \left(-1 + \frac{x_k}{\lambda} \right).$$

- MLE $\hat{\lambda}$ makes $\frac{d\ell(\hat{\lambda})}{d\lambda} = 0$.

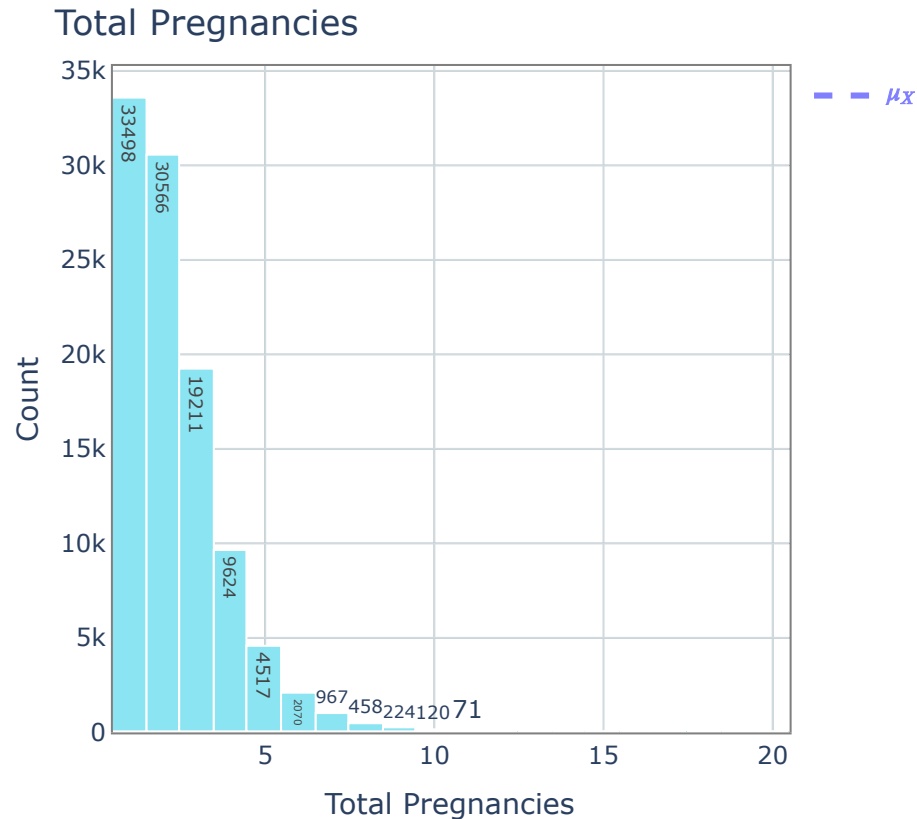
$$\Leftrightarrow \sum_{k=1}^n \left(-1 + \frac{x_k}{\hat{\lambda}} \right) = 0$$

$$\Leftrightarrow \hat{\lambda} = \frac{1}{n} \sum_{k=1}^n x_k.$$

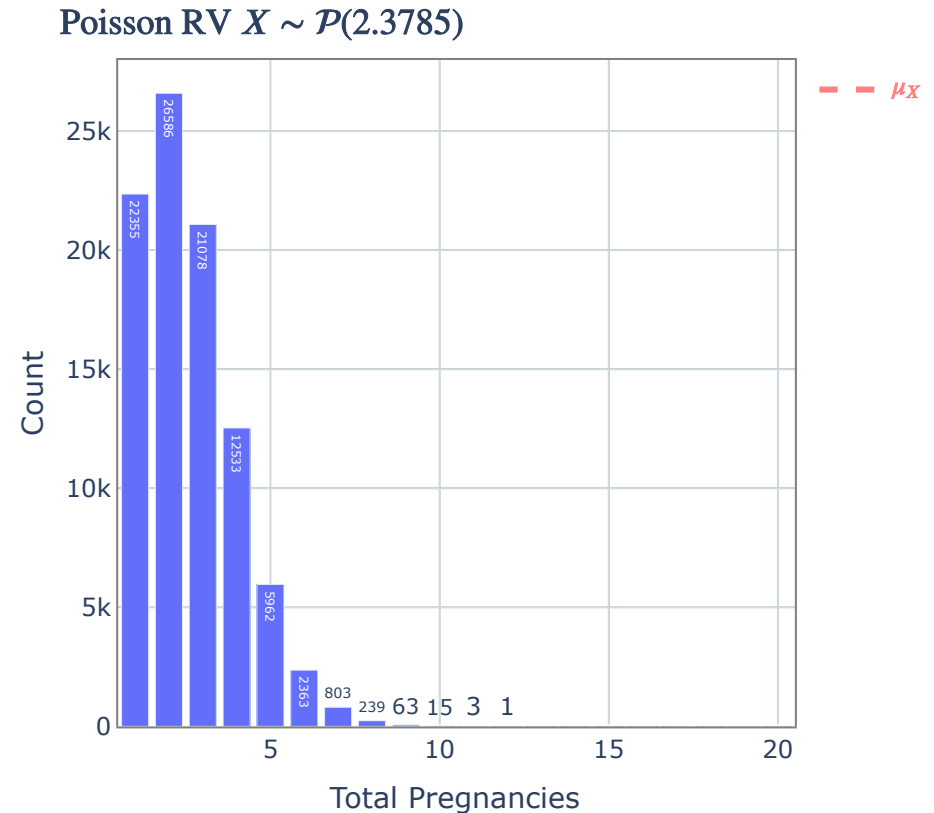
- MLE $\hat{\lambda} = \bar{X}_n = 2.38$.

Data Vs Theoretical Distribution

Data



Simulation



- The choice of your model matters!
- How about trying $X \sim \mathcal{E}(\lambda)$?

4 Simulation Methods

4.1 Uniform Random Variable

Linear Congruential Generator (LCG)

- **Fact** We humans cannot produce real randomness!
- Random numbers produced by computer are called **pseudorandom numbers**.
- **LCG** is a method to generate pseudorandom uniform numbers by generating a sequence of large integers and then mapping them to the interval $[0, 1]$.
- **Park-Miller LCG (1988)** chose $a = 7^5$, $b = 0$, and $m = 2^{31} - 1$:
 - Set seed $X_0 \in \{0, 1, \dots, m - 1\}$, then generate $u_{n+1} = \frac{X_{n+1}}{m}$,
 - $X_{n+1} = (a \cdot X_n + b) \bmod m$.

4.2 Other Distributions

Bernoulli Distribution

- **Standard Uniform Distribution** allows us to generate almost every other common distributions.
- **Bernoulli Distribution** can be generated very easily from a uniform distribution.
- Let p be the probability of success, then:
 - Generate a random number $u \sim U[0, 1]$,
 - If $u < p$, return $X = 1$ (success),
 - Otherwise, return $X = 0$ (failure).
- Check that $X \sim \mathcal{B}(p)$.

Binomial Distribution

- From **Bernoulli Distribution**, we can generate a **Binomial Distribution** by summing up n independent Bernoulli trials:
 - Initialization: n, p
 - We generate n trials of $X_i \sim \mathcal{B}(p)$, let $Y = \sum_{i=1}^n X_i$
- Check that $Y \sim \mathcal{B}(n, p)$.

Number of eggs (n)

10

Hatching probability (p)

0.5

▶ Run Simulation



Number of chicks: 0

Exponential Distribution

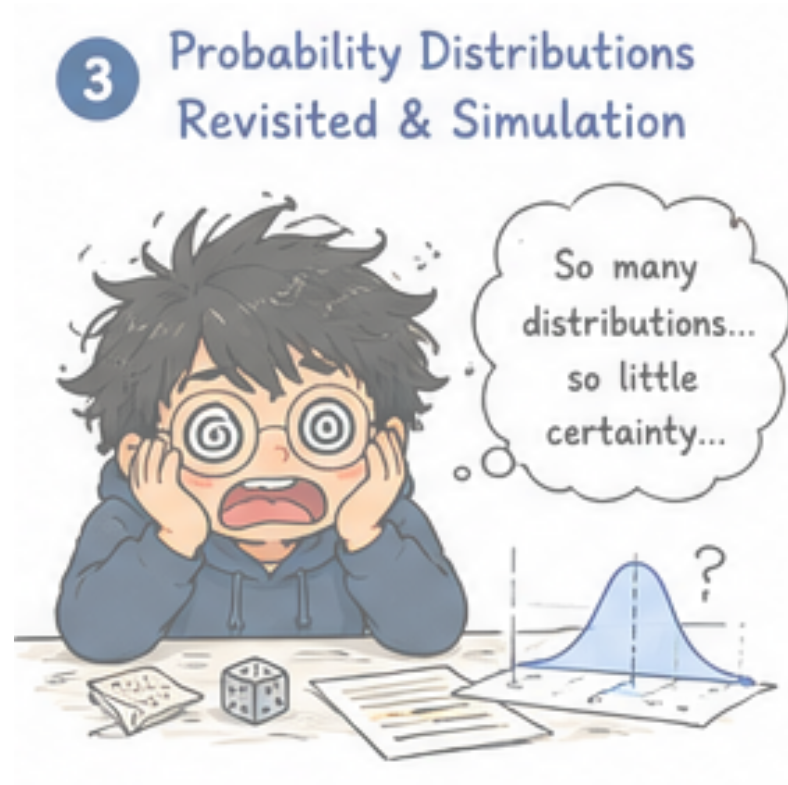
- From **Uniform Distribution**, we can generate an **Exponential Distribution** using **Inverse Method**.
- **Definition:** Cumulative Distribution Function (CDF) of a random variable X is a function $F(x) = \mathbb{P}(X \leq x)$.
 - If X is discrete, $F(x) = \sum_{x_i \leq x} \mathbb{P}(X = x_i)$, the sum of **PMF**.
 - If X is continuous, $F(x) = \int_{-\infty}^x f(x)dx$, where f is its **PDF**.
- **Inverse Transform Sampling:** if X is a RV with CDF $F(x)$, and $U \sim U(0, 1)$, then $Y = F^{-1}(U)$ has the same distribution as X .
- **Apply to Exponential Distribution**
 - Simulate $U \sim U(0, 1)$,
 - Compute $X = -\frac{1}{\lambda} \log(1 - U)$, then $X \sim \mathcal{E}(\lambda)$.

Poisson Distribution

- From **Exponential Distribution** of parameter λ , we can generate a **Poisson RV** with parameter λ by counting the number of events occur before the total times is 1.
- **Simulation:**
 - Simulate $X_1, X_2, \dots \sim \mathcal{E}(\lambda)$,
 - Let $S_n = X_1 + X_2 + \dots + X_n$,
 - Let $N = \max\{n | S_n \leq 1\}$, then $N \sim \mathcal{P}(\lambda)$.

```
1 import numpy as np
2 def poisson_from_exponential(lam):
3     t = 0
4     n = 0
5     while True:
6         t += np.random.exponential(1 / lam)
7         if t > 1:
8             return n
9         n += 1
```

🥳 Yeahhhh 🍷 !!!



Let's take a break!